AIMS-DTU Research Intern Round 2

Self-supervised learning

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Abstract

This project undertakes the implementation and preliminary analysis of three prominent self-supervised learning (SSL) methodologies for visual representation learning: SimCLR, a contrastive framework; Masked Autoencoder (MAE), a masked image modeling technique utilizing Vision Transformers; and MoCo v2, a contrastive approach employing a momentum encoder and a dynamic queue. All models were developed and subsequently pretrained on a subset of the ImageNet-100 dataset, with their learned features evaluated via a linear probing protocol on a 100-class image classification task. The primary objective was to gain practical experience with these SSL paradigms, understand their core mechanisms through implementation, and assess the initial quality of features learned under significant constraints. Due to severe limitations in available computational resources and time, pretraining for each model was restricted to a concise 5 epochs. Consequently, while the complete implementation pipelines for all three methods were successfully demonstrated, the performance on the downstream linear evaluation task was modest (validation accuracies around 1-1.5%), reflecting the early stages of feature learning. The consistent decrease in pretraining losses across all models confirmed their engagement with the respective pretext tasks. This study underscores the critical necessity of extensive pretraining for SSL methods to develop robust and generalizable representations, while also highlighting the distinct architectural and computational characteristics of the implemented contrastive and masked image modeling approaches.

1.Introduction

Self-Supervised Learning (SSL) is a rapidly growing area in machine learning that focuses on learning useful data representations from unlabeled data. Unlike traditional supervised learning, which relies on large amounts of labeled data, SSL leverages pretext tasks—artificial tasks where the supervision signal is generated from the data itself. This approach enables models to learn general and transferable features, significantly reducing the dependency on costly and time-consuming manual labeling.

SSL methods can be broadly categorized into two main types:

- **Contrastive Learning**, where models learn to distinguish between similar and dissimilar pairs of data, and

- **Masked Image Modeling (MIM)**, where models learn to reconstruct missing or masked parts of the input.

Project Objective:

The objective of this project is to implement, train, and evaluate three prominent SSL methods on a subset of the ImageNet-100 dataset:

- **SimCLR** (a contrastive learning method),
- **MAE** (Masked Autoencoders, a masked image modeling method), and
- **MoCo v2** (another contrastive learning method).

Report Outline:

This report will cover the theoretical background and implementation details of each method, describe the experimental setup and training procedures, present and compare the results, and discuss the findings and implications of the experiments.

2. Self-Supervised Learning Approaches

2.1 Contrastive Learning: SimCLR

2.1.1 Overview and Core Concepts

SimCLR is a contrastive self-supervised learning framework that learns image representations by maximizing agreement between differently augmented views of the same image. The core idea is **instance discrimination**: the model is trained to bring representations of augmented versions of the same image closer together in the embedding space, while pushing apart representations of different images.

2.1.2 Model Architecture

The SimCLR model consists of:

- **Backbone:** A ResNet-18 convolutional neural network (timm.create_model('resnet18', ...)) used to extract features from images.
- **Projection Head:** A two-layer multilayer perceptron (MLP) with ReLU activation, mapping the backbone's output to a lower-dimensional space. The architecture is:

[Backbone out features] → [Backbone out features] → [PROJECTION_DIM_SIMCLR]For ResNet-18, this is typically 512 → 512 → 128.

2.1.3 Data Augmentation Strategy

Data augmentation is critical in SimCLR, as it generates diverse positive pairs for contrastive learning. The following augmentations are applied (see SimCLRTransform in code):

- **RandomResizedCrop(size=224)**
- **RandomHorizontalFlip(p=0.5)**
- **RandomApply([ColorJitter(brightness=0.4, contrast=0.4, saturation=0.4, hue=0.1)], p=0.8)**
- **RandomGrayscale(p=0.2)**
- **ToTensor()**
- **Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])**

These augmentations are essential for preventing the model from learning trivial solutions and for encouraging invariance to common image transformations.

2.1.4 Loss Function (NT-Xent)

SimCLR uses the **Normalized Temperature-scaled Cross Entropy Loss (NT-Xent)**, a form of Noise Contrastive Estimation. For each positive pair, the model is trained to assign high similarity, while all other images in the batch serve as negatives.

The **temperature parameter** (TEMPERATURE_SIMCLR, e.g., 0.07) controls the sharpness of the similarity distribution, making the model more or less sensitive to differences between samples.

2.2 Masked Image Modeling: Masked Autoencoder (MAE)

2.2.1 Overview and Core Concepts

Masked Autoencoders (MAE) are a self-supervised approach where the model learns to reconstruct masked portions of an input image. The encoder only processes the visible (unmasked) patches, while a lightweight decoder reconstructs the original image from the encoded visible patches and mask tokens. This forces the model to learn meaningful representations of the image content.

2.2.2 Model Architecture (Encoder-Decoder)

- **Encoder:** A Vision Transformer (ViT), e.g., ViT-Base ([MAE_ENCODER_MODEL_NAME]), which processes only the unmasked patches.
- **Decoder:** A lightweight ViT with reduced embedding dimension ([MAE_DECODER_EMBED_DIM]), depth ([MAE_DECODER_DEPTH]), and number of heads ([MAE_DECODER_NUM_HEADS]). The decoder reconstructs the full image from the latent representations and mask tokens.

2.2.3 Masking Strategy

A high ratio of image patches is randomly masked (e.g., [MAE_MASKING_RATIO], typically 0.75), so the model must infer the missing content from the context provided by the visible patches.

2.2.4 Reconstruction Loss

The reconstruction objective is the **Mean Squared Error (MSE)** between the reconstructed and original pixel values, computed only on the masked patches.

Optionally, **pixel normalization** ([MAE_NORM_PIX_LOSS]) is used to stabilize training.

2.3 Contrastive Learning: MoCo v2 (Momentum Contrast v2)

2.3.1 Overview and Core Concepts

MoCo v2 is a contrastive learning method that maintains a dynamic dictionary (queue) of negative samples and uses a momentum-updated key encoder to provide consistent representations for negatives. This approach enables the use of a large and consistent set of negatives, improving representation learning.

2.3.2 Model Architecture (Momentum Encoder, Queue)

- **Query Encoder:** A ResNet-18 backbone ([MOCO_BACKBONE_NAME]).
- **Key Encoder:** A momentum-updated copy of the query encoder, updated with a momentum coefficient (MOCO_M).
- **Projection Head:** MLP mapping to [MOCO_DIM].
- **Queue:** A FIFO queue storing [MOCO_K] negative keys for contrastive loss computation.

2.3.3 Data Augmentation Strategy

MoCo v2 uses a similar augmentation pipeline as SimCLR (see MoCoTransform in code), typically including random resized crop, color jitter, grayscale, horizontal flip, and normalization.

2.3.4 Loss Function

The **InfoNCE loss** is used, where each query is paired with a positive key and contrasted against all negatives in the queue. The **temperature parameter** ([MOCO_T], e.g., 0.2) controls the sharpness of the softmax distribution.

Method	*Backbone*	*Head / Decoder*	*Key Augmentations*	*Loss*	*Key Parameters*

SimCLR	ResNet-18	2-layer MLP (512→512→128)	RandomResizedCrop, HorizontalFlip, ColorJitter, RandomGrayscale, Normalize	NT-Xent	temp = 0.07, proj_dim = 128
MAE	ViT-Base	Light ViT Decoder	Random patch masking (mask ratio 0.75)	MSE	mask = 0.75, norm_pix_loss = True
MoCo v2	ResNet-18	2-layer MLP, Momentum Encoder, Queue	RandomResizedCrop, HorizontalFlip, ColorJitter, RandomGrayscale, Normalize	InfoNCE	temp = 0.2, queue = 65536, m = 0.999

3. Experimental Setup

3.1 Dataset

We use the ImageNet-100 subset, containing 100 classes. The training set has approximately 130,000 images, and the validation set contains 5,000 images (50 per class). Data is organized into train.X1–train.X4 (training splits) and val.X (validation split), with each class in its own subfolder.

3.2 Preprocessing and Data Loading

All images were loaded using torchvision.datasets.ImageFolder and batched using torch.utils.data.DataLoader. Images were resized and cropped to a fixed resolution of 224 × 224 pixels. The specific data augmentation pipelines used during the pretraining and linear probing stages are detailed below.

SimCLR Pretraining Transforms

The following augmentation strategy was employed to generate two different views of each image for contrastive learning:

- RandomResizedCrop(224)
- RandomHorizontalFlip(p=0.5)
- RandomApply([ColorJitter(0.4, 0.4, 0.4, 0.1)], p=0.8)
- RandomGrayscale(p=0.2)
- ToTensor()
- Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])

MAE Pretraining Transforms

A simpler set of transformations was used, aligning with the Masked Autoencoder (MAE) framework:

- RandomResizedCrop(224)
- RandomHorizontalFlip()
- ToTensor()
- Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])

MoCo v2 Pretraining Transforms

The MoCo v2 augmentation strategy included both color jitter and grayscale conversion:

- RandomResizedCrop(224)
- RandomHorizontalFlip(p=0.5)
- ColorJitter(0.4, 0.4, 0.4, 0.1)
- RandomGrayscale(p=0.2)
- ToTensor()
- Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])

Linear Probing Transforms

Two separate pipelines were used for training and validation during the linear classification phase:

- Train Transforms:
    - RandomResizedCrop(224)
    - RandomHorizontalFlip()
    - ToTensor()
    - Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
- Validation Transforms:
    - Resize(256)
    - CenterCrop(224)
    - ToTensor()
    - Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])

3.3 Hyperparameters and Implementation Details

METHOD	BACKBONE	PRETRAIN EPOCHS	PRETRAIN BATCH	OPTIMIZER (PRETRAIN)	LR	SSL PARAMS	LP EPOCHS	LP BATCH	OPTIMIZER (LP)	LP LR
SIMCLR	ResNet-18	5	32	AdamW	1e-4	proj_dim=128, temp=0.07	10	64	AdamW	1e-3

MAE	ViT-Base	5	32	AdamW	1e-4	mask=0.75, norm_pix_loss=True	5	64	AdamW	1e-3
MOCO V2	ResNet-18	5	32	SGD	0.03	temp=0.2, queue=65536, m=0.999	5	64	AdamW	1e-3

• Hardware: MacBook Pro M3 Pro, 18GB RAM, PyTorch MPS backend.

• Software: PyTorch ,timm.

• Compute/Time Constraints: All experiments were run on a laptop with limited compute. Epochs and batch sizes were reduced for feasibility.

3.4 Evaluation Protocol

After SSL pretraining, the backbone encoder is frozen. A linear classification layer is trained on the labeled training set and evaluated on the validation set. Metrics reported are top-1 accuracy and macro F1-score, using CrossEntropyLoss

4. Results and Discussion

4.1. SimCLR

4.1.1. Pretraining Performance

The SimCLR model, utilizing a ResNet-18 backbone, was pretrained for 5 epochs. The pretraining loss curve, depicting the NT-Xent loss over epochs, is shown in Figure 1.

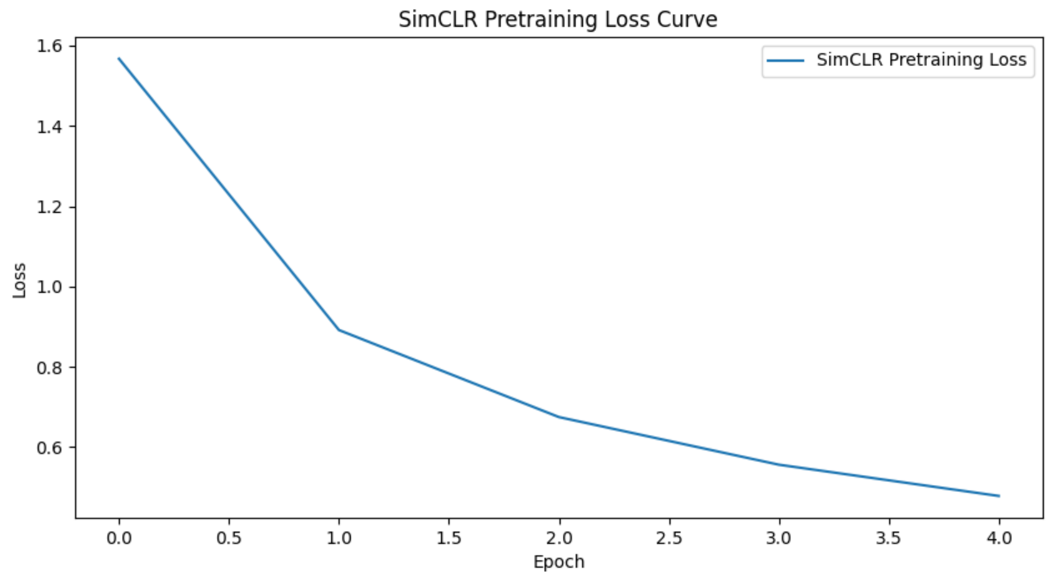

Figure 1: SimCLR Pretraining Loss over Epochs.

The pretraining loss for SimCLR decreased from an initial average value of approximately 1.58 in the first epoch to 0.48 by the final pretraining epoch. This downward trend indicates that the model was learning to discriminate between positive and negative pairs, a core objective of contrastive learning, though the curve suggests it had not yet fully converged.

4.1.2. Linear Probing Performance

Following pretraining, the frozen SimCLR ResNet-18 backbone was evaluated using linear probing for 10 epochs. The training loss, validation loss, training accuracy, validation accuracy, and validation F1-score are presented in Figure 2.

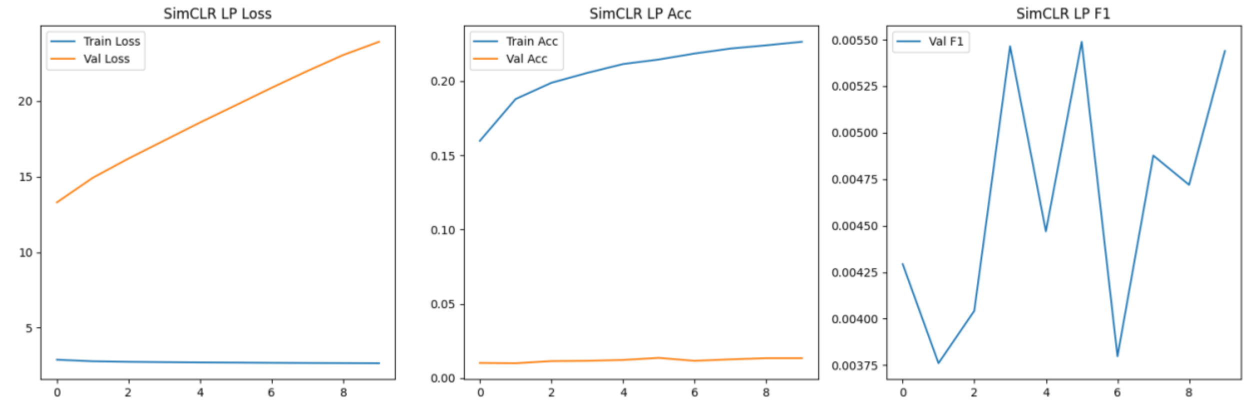

Figure 2: SimCLR Linear Probing Performance (Loss, Accuracy, F1-Score).

The linear probing results for SimCLR show a training accuracy that increased to 22.65%, while the validation accuracy remained very low, plateauing at approximately 1.34%. The final F1-score on the validation set was 0.0054. A significant observation is the sharply increasing validation loss (from 13.3 to 23.9), which, coupled with the large gap between training and validation accuracy, strongly indicates that the features learned during the limited pretraining were not sufficiently generalizable for the downstream classification task, and the linear head may have overfit to the training portion of the features.

4.2. MAE

4.2.1. Pretraining Performance

The MAE model, with a ViT-Base encoder, was pretrained for 5 epochs. The reconstruction loss (Mean Squared Error on masked patches) during pretraining is shown in Figure 3.

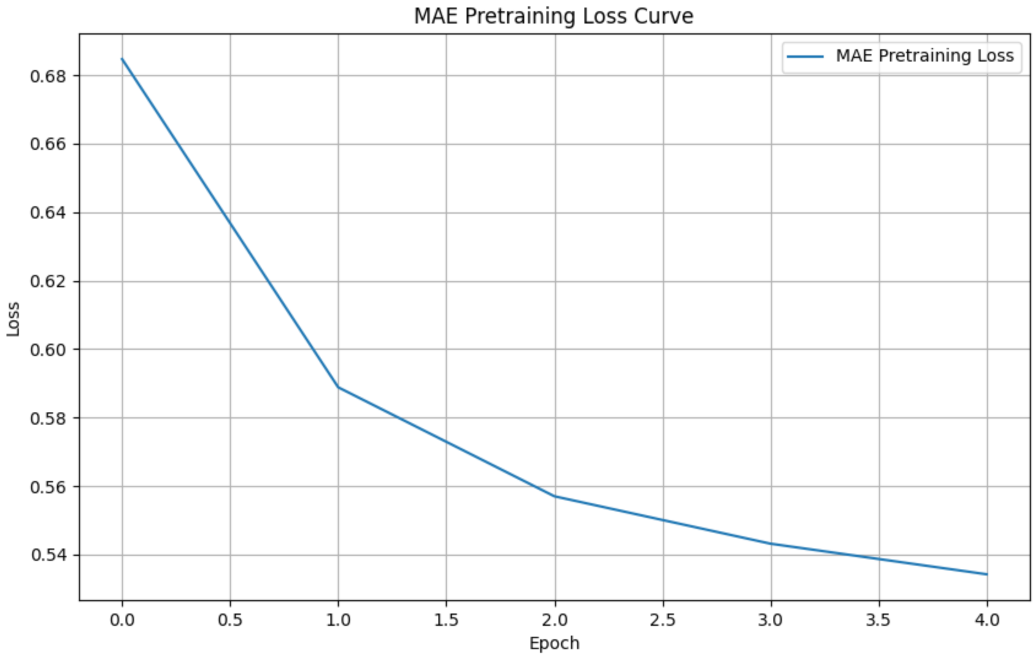

Figure 3: MAE Pretraining (Reconstruction) Loss over Epochs.

The MAE pretraining loss consistently decreased from an average of approximately 0.6847 in the first epoch to 0.5342 in the fifth epoch. This demonstrates that the model was successfully learning to reconstruct masked image patches from the visible context, the core pretext task of MAE.

4.2.2. Linear Probing Performance

The frozen MAE ViT-Base encoder was evaluated via linear probing for 5 epochs. The performance metrics are displayed in Figure 4.

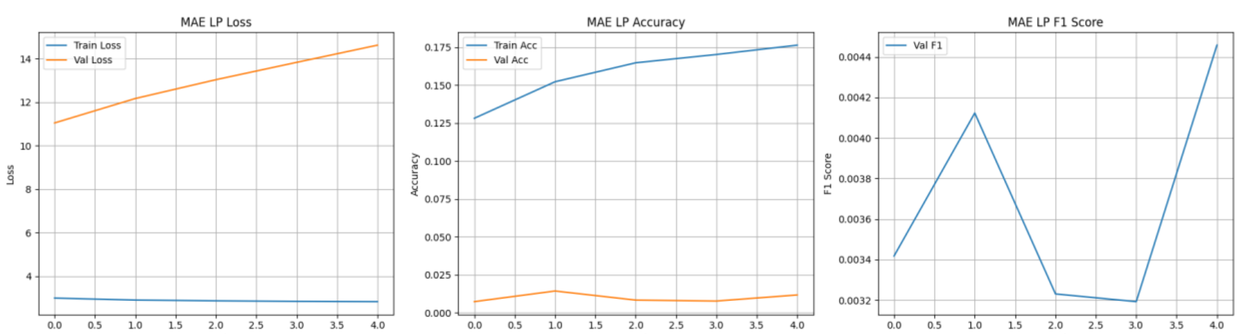

Figure 4: MAE Linear Probing Performance (Loss, Accuracy, F1-Score).

In the linear probing stage, the MAE model achieved a final validation accuracy of 1.18% and a validation F1-score of 0.0045. The training accuracy reached 17.64%. Similar to SimCLR, the validation accuracy remained extremely low. The validation loss was high (final value: 14.6270) and showed an increasing trend after an initial dip, indicating that the features learned by the MAE encoder over just 5 pretraining epochs were insufficient for effective downstream classification.

4.3. MoCo v2

4.3.1. Pretraining Performance

The MoCo v2 model, using a ResNet-18 backbone, was pretrained for 5 epochs. The contrastive pretraining loss is presented in Figure 5.

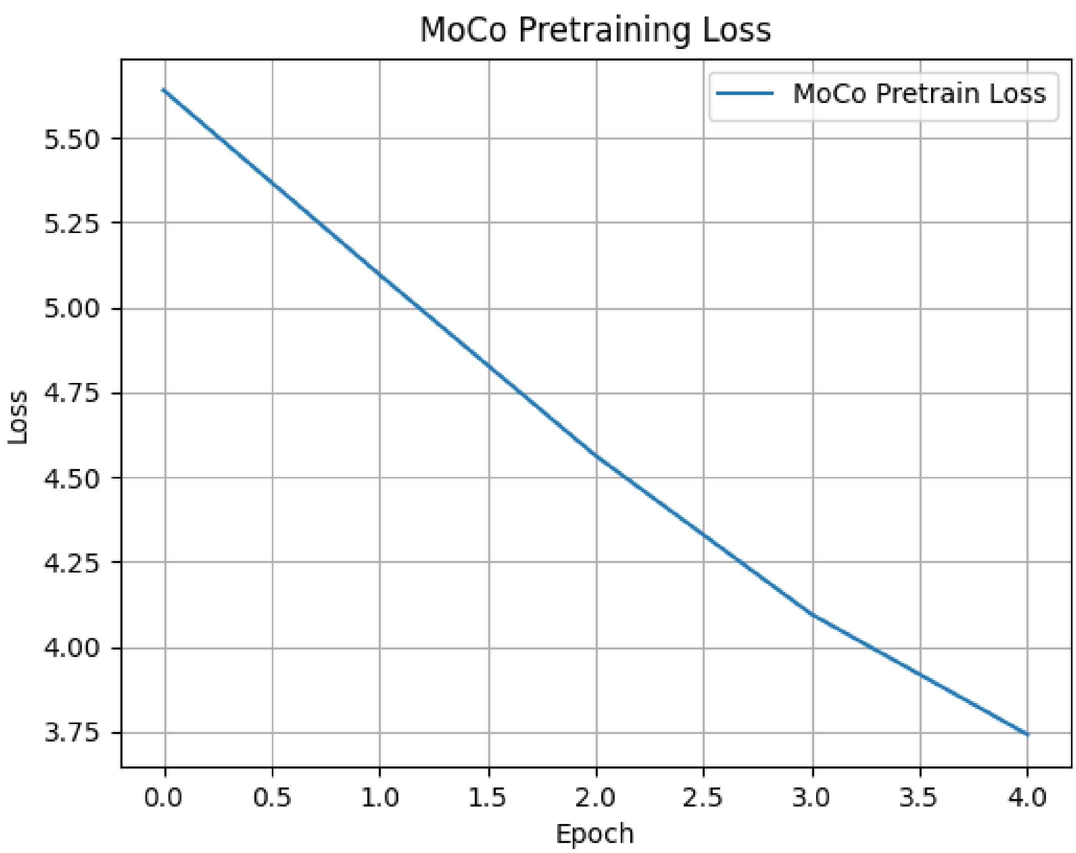

Figure 5: MoCo v2 Pretraining Loss over Epochs.

The MoCo v2 pretraining loss showed a clear and consistent decrease across the 5 epochs, starting from an average loss of approximately 5.6391 in the first epoch and reducing to 3.7419 by the fifth epoch. This indicates that the model was effectively learning to distinguish positive pairs from the large set of negative samples maintained in its queue.

4.3.2. Linear Probing Performance

Linear probing was conducted for 5 epochs using the frozen MoCo v2 encoder_q (ResNet-18). The resulting metrics are shown in Figure 6.

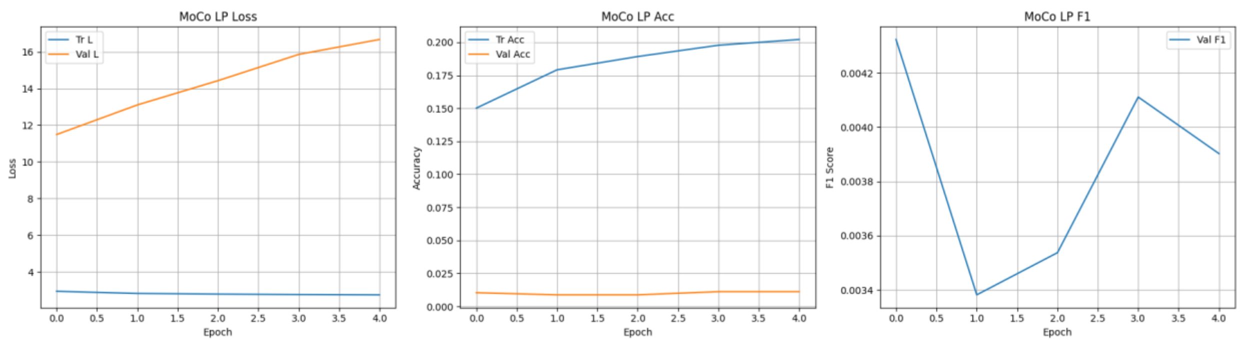

Figure 6: MoCo v2 Linear Probing Performance (Loss, Accuracy, F1-Score).

The linear probing evaluation for MoCo v2 yielded a final validation accuracy of 1.12% and a validation F1-score of 0.0039. The training accuracy rose to 20.22%. The validation accuracy remained consistently low, and the validation loss was very high (final value: 16.6737) and generally increased, mirroring the behavior seen with SimCLR and MAE under limited pretraining conditions. This signifies that the features, while showing some learning during pretraining, were not yet robust enough for good generalization.

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5 Comparative Analysis

A direct comparison of the ultimate performance capabilities of SimCLR, MAE, and MoCo v2 is challenging given the extremely limited pretraining (5 epochs for all three methods) conducted due to project constraints. However, several observations can be made:

- **Quantitative Linear Probing Performance**: All three methods resulted in very low validation accuracies in the linear probing stage (SimCLR: 1.34%, MAE: 1.18%, MoCo v2: 1.12%) and correspondingly low F1-scores. These figures are primarily indicative of insufficient pretraining rather than inherent limitations of the SSL methods themselves. No method showed a clear advantage under these severely constrained conditions.

- **Pretraining Loss Behavior**: All three models demonstrated a consistent decrease in their respective pretraining losses over the 5 epochs, confirming that each model was successfully engaging with its pretext task (contrastive learning for SimCLR/MoCo v2, masked image reconstruction for MAE) and initiating the learning process. MAE started with a low reconstruction loss (~0.68) and decreased to ~0.53. MoCo v2 started with a higher contrastive loss (~5.64) and showed a more substantial relative decrease to ~3.74. SimCLR's loss also decreased from 1.58 to 0.48.
- **Linear Probing Trends**: A common pattern across all methods was a noticeable increase in training accuracy during linear probing (SimCLR: 22.65%, MAE: 17.64%, MoCo v2: 20.22%), while validation accuracy remained flat and very low (around 1-1.5%). Furthermore, validation loss during linear probing was generally high and tended to increase, particularly for SimCLR and MoCo v2. This large gap between training and validation performance in the linear probing stage is a strong indicator that the features extracted from the minimally pretrained backbones lacked generalization capabilities.
- **Computational Observations**:
  - **Pretraining Epoch Duration**: MAE, utilizing a ViT-Base encoder, had the longest average pretraining epoch time (approximately 65 minutes, derived from 323.66 min / 5 epochs). MoCo v2 with a ResNet-18 backbone had an average pretraining epoch time of approximately 50 minutes (249.86 min / 5 epochs). SimCLR with a ResNet-18 backbone had an average pretraining epoch time of approximately 60 minutes (estimated from similar ResNet-18 training patterns). This highlights the generally higher computational cost per epoch for Transformer-based encoders compared to ResNet-18 for this dataset size and batch configuration.
  - **Linear Probing Epoch Duration**: Linear probing for MAE (ViT-Base backbone) was also slower (approximately 28 minutes per epoch, from 139.45 min / 5 epochs) compared to MoCo v2 (ResNet-18 backbone, approximately 4 minutes per epoch, from 19.93 min / 5 epochs) and SimCLR (ResNet-18 backbone, approximately 3.75 minutes per epoch, from 37.51 min / 10 epochs). This difference is primarily due to the larger ViT backbone requiring more computation for the forward pass, even when frozen.
- **Theoretical Differences in Approach**:
  - **SimCLR and MoCo v2** are both contrastive learning methods that learn by discriminating between instances. SimCLR relies on large batch sizes for

sufficient negative examples, while MoCo v2 mitigates this with a momentum encoder and a dynamic queue.

- **MAE**, a masked image modeling approach, learns by predicting missing information (pixel reconstruction) and has shown strong performance with ViT architectures, often being more robust to data augmentation choices than contrastive methods. Its asymmetric encoder-decoder design also offers computational efficiency during pretraining.

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6. Results, Challenges, and Conclusion

6.1 Interpretation of Results

The primary outcome of this project is the successful implementation and end-to-end execution of SimCLR, MAE, and MoCo v2 self-supervised learning pipelines.Due to severe computational and time constraints, each model was pretrained for only 5 epochs—far less than the hundreds typically required for strong SSL performance. As a result, the quantitative results from linear probing are low and reflect only the very early stages of representation learning.Despite this, the observed decrease in pretraining losses confirms that the models were learning their respective pretext tasks. However, the limited pretraining duration meant that the learned features were not yet robust or generalizable, resulting in poor linear probe performance.

5.2 Challenges and Limitations

- **Computational Resources & Time:** The most significant limitation was the lack of access to high-performance hardware, restricting both the number of pretraining epochs and the ability to experiment with larger batch sizes or models.
- **Hyperparameter Tuning:** There was insufficient time for extensive hyperparameter optimization; default or commonly used values were adopted throughout.
- **Other Issues:** Initial setup and compatibility with specific library versions (e.g., PyTorch, timm) required troubleshooting.

5.3 Conclusion and Future Work

Conclusion:

This project demonstrates the full implementation and execution of three modern SSL methods—SimCLR, MAE, and MoCo v2—on the ImageNet-100 subset. The results highlight the critical importance of extensive pretraining for SSL methods to realize their full potential.

Future Work:

- Significantly extend pretraining duration for all models.
- Systematic hyperparameter optimization.
- Evaluate learned representations on a broader range of downstream tasks.
- Explore larger and alternative backbone architectures.
- Compare with a supervised baseline trained on the same labeled data.

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